

A PROJECT REPORT ON

**POWER FACTOR CORRECTION (PFC) FOR
INDUSTRIAL POWER USE TO MINIMIZE
PENALTY**

Submitted as semester project

For the ENA course

SUBMITTED BY

----Ammar Asad Warraich----

---Ali Jarrar---

---Rana Hamza---

---Uzair Chaudhary---

---Waqas---

DEPARTMENT OF ELECTRICAL ENGINEERING
SEEDS-SCHOOL OF ELECTRICAL ENGINEERING
AND COMPUTER SCIENCE ,NUST UNIVERSITY

ACKNOWLEDGEMENT

We take pleasure in presenting before you, our project, which is result of studied blend of both practice and knowledge.

We express our earnest gratitude to our Lab Engineer, Madam Tehniyat Siddiqui , Department of EE, our project guide, for her encouragement and guidance.

Finally, we express our gratitude to all other members who are involved either directly or indirectly for the completion of this project.

CONTENTS

- 1. ABSTRACT**
- 2. POWER FACTOR IMPROVEMENT**
- 3. INTRODUCTION TO EMBEDDED SYSTEMS**
- 4. BLOCK DIAGRAM**
- 5. HARDWARE REQUIREMENTS**
 - 4.1 TRANSFORMERS**
 - 4.2 VOLTAGE REGULATOR (LM7805)**
 - 4.3 RECTIFIER**
 - 4.4 FILTER**
 - 4.5 MICROCONTROLLER (AT89S52/C51)**
 - 4.6 RELAY**
 - 4.7 RELAY DRIVER**
 - 4.8 PUSH BUTTONS**
 - 4.9 LCD**
 - 4.10 LM339**
 - 4.11 CURRENT TRANSFORMER**
 - 4.12 INDUCTIVE LOAD**
 - 4.13 SHUNT CAPACITOR**
 - 4.14 RESISTIVE LOAD**
 - 4.15 LED**
 - 4.16 1N4008 / 1N4148**

4.17 RESISTOR

4.18 CAPACITOR

5. BIBLIOGRAPHY

CHAPTER 1

INTRODUCTION

OBJECTIVE

PROBLEM STATEMENT

INTRODUCTION

With everything moving toward new tech these days, it's pretty clear that power is one of our most valuable resources. In industrial setups, we're seeing more inductive loads being used every day, which is a problem because it drags down the system's power factor and makes everything less efficient. This leads to a lagging power factor, and for consumers, that usually means getting hit with high penalties on their electricity bills.

We deal with these penalties using Power Factor Control. Basically, it's a way to fix the negative effects of loads that pull the power factor below 1.0. In AC circuits, you can think of the power factor as the ratio between the real power (the stuff actually doing the work) and the apparent power being supplied to the circuit. Real power is just the circuit's capacity to do work in a certain amount of time.

This project focuses on a way to provide continuous power factor correction automatically, so we don't have to manually load capacitor banks. By using Power Factor Correction Capacitors, we can handle the reactive power created by a load which is what utility companies actually target when they see inefficient energy use on a bill. Adding these capacitors is a standard way to reduce the load, cut down on wasted energy, and ultimately lower those electricity costs.

Power factor improvement

Alternating current circuits

Unlike Direct Current Circuits, where only resistance restricts the current flow, in Alternating Current Circuits, there are other circuit aspects which determine the current flow; though these are akin to resistance, they do not consume power, but load the system with reactive currents; like D.C. circuits where the current multiplied by voltage gives watts, here the same gives only VA.

Like resistance, these are called “Reactance”. Reactance is caused by either inductance or by capacitance. The current drawn by inductance lags the voltage while the one by capacitance leads the voltage. Almost all industrial loads are inductive in nature and hence draw lagging wattless current, which unnecessarily loads the system, performing no work. Since the capacitive current is leading in nature, loading the system with capacitors wipes out them.

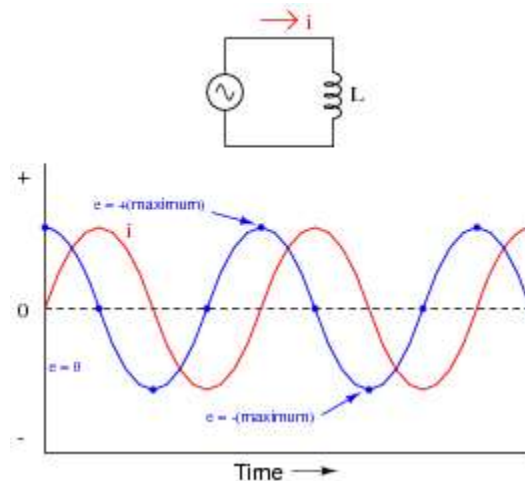


FIG: WAVEFORMS FOR INDUCTIVE LOAD

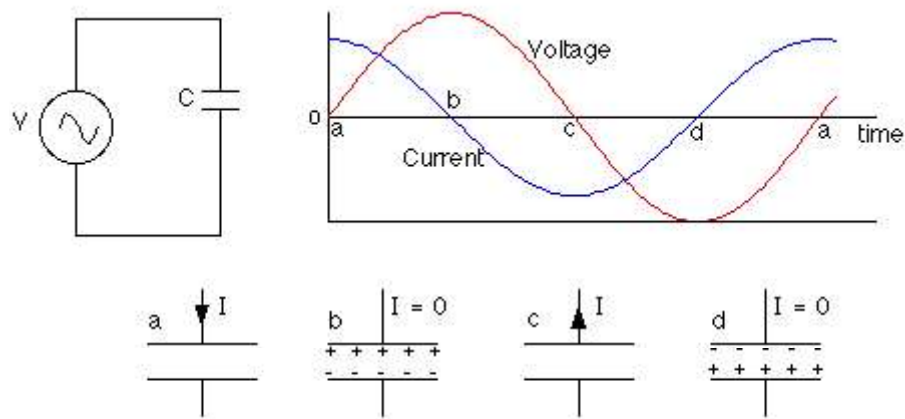


FIG: WAVEFORMS FOR CAPACITIVE LOAD

Capacitors for power-factor improvement

Basically, no matter what the power factor ends up being, the power company has to install equipment that can handle a specific amount of voltage and current—even if a lot of that isn't actually doing any useful work. The generators have to be built to withstand their rated capacity regardless of the actual power being delivered.

For example, if you have an alternator rated for 1000A at 11kV, the coils have to be able to carry that full current. The apparent power for that machine is 11 MVA. If the load power factor is 1.0 (unity), then all 11 MVA is delivered as 11 MW of active power, which means the alternator is running at its full potential.

But, if the power factor drops to 0.8 lagging, the consumer is only using 8.8 MW. The power company only gets paid for that 8.8 MW, but they still have to run a generator rated for the full 1000A and 11kV. From the supplier's perspective, the lower the power factor gets, the worse the deal is for them. Because of this, they push consumers to fix their power factor and usually hand out penalties if they don't. Improving it just comes down to reducing the phase angle between the supply voltage and the current.

Location of power-factor improvement capacitor banks:

Any installation including the following types of machinery or equipment is likely to have low power factor which can be corrected, with a consequent saving in charges, by way of reduced demand charges, lesser low power factor penalties:

1. Induction motors of all types
2. Power thyristor installation
3. Power transformers and voltage regulators.
4. Welding machines
5. Electric-arc and induction furnaces.
6. Choke coils and magnetic system.
7. Neon signs and fluorescent lighting.

Apart from penalties like maximum demand charges, penalty for low power factor, the factory cabling and supply equipment can be relieved of a considerable wattles or reactive load, which will enable additional machinery to be connected to the supply without enlarging these services. Additionally, the voltage drop in the system is reduced.

The method employed to achieve the improvements outlined involves introducing reactive kVA (kvar) into the system in phase opposition to the wattles or reactive current mentioned above the effectively cancels its effect in the system. This is achieved either with rotary machines (synchronous condensers)

POWER FACTOR CORRECTION

The power factor of an [AC](#) electric power system is defined as the [ratio](#) of the [real power](#) flowing to the load to the [apparent power](#) in the circuit, and is a dimensionless number between 0 and 1 (frequently expressed as a percentage, e.g. 0.5 pf = 50% pf). Real [power](#) is the capacity of the circuit for performing work in a particular time. Apparent power is the product of the current and voltage of the circuit. Due to energy stored in the load and returned to the source, or due to a non-linear load that distorts the wave shape of the current drawn from the source, the apparent power will be greater than the real power.

In an electric power system, a load with a low power factor draws more current than a load with a high power factor for the same amount of useful power transferred. The higher currents increase the energy lost in the distribution system, and require larger wires and other equipment. Because of the costs of larger equipment and wasted energy, electrical utilities will usually charge a higher cost to industrial or commercial customers where there is a low power factor.

Linear loads with low power factor (such as [induction motors](#)) can be corrected with a passive network of [capacitors](#) or [inductors](#). Non-linear loads, such as [rectifiers](#), distort the current drawn from the system. In such cases, active or passive power factor correction may be used to counteract the distortion and raise the power factor. The devices for correction of the power factor may be at a central substation, spread out over a distribution system, or built into power-consuming equipment.

This is a lot of great detail, but definitely sounds like it was pulled straight from a textbook or a white paper. I've rewritten it to sound more like a 2nd-

semester EEE student's report—technical and accurate, but using the kind of language you'd actually use in a lab or project documentation.

Power Factor in Linear Circuits

In a simple resistive AC circuit, the voltage and current waveforms are "in step" (in phase). They change polarity at the exact same time, and all the power going into the load is actually used. However, when we add reactive loads like inductors or capacitors energy gets stored in the load, which creates a time delay between the current and voltage waves.

During each AC cycle, extra energy (on top of what the load actually consumes) is temporarily stored in magnetic or electric fields and then dumped back into the grid a fraction of a second later. This "ebb and flow" doesn't do any real work, but it increases the total current in the line. This means a circuit with a low power factor needs way more current to move the same amount of real power compared to a high power factor circuit. Linear loads don't mess with the wave shape, but they definitely shift the timing (the phase) between V and I .

Anything with a simple heating element (like a toaster or a filament bulb) has a power factor of 1.0. But things like motors or solenoid valves usually have a power factor below 1.0 because of their inductive components.

Definitions and Math

In AC power, we look at three main components:

1. **Real Power (P):** Measured in Watts (W) this is the power actually doing work.
2. **Apparent Power (S):** Measured in Volt-Amperes (VA) the total power supplied.

3. **Reactive Power (Q):** Measured in Volt-Amperes Reactive (VAR) the power bouncing back and forth.

If the waveform is a perfect sine wave, we can represent these as a vector triangle. If θ is the phase angle between V and I , then the power factor is just $\cos(\theta)$.

Since the units are consistent, the power factor is a dimensionless number between 0 and 1. If it's 0, the energy is entirely reactive (just moving back and forth). If it's 1, the load is consuming everything. We usually label these as "leading" or "lagging."

- **Lagging:** Inductive loads (like motors or transformers) cause the current to lag behind the voltage.
- **Leading:** Capacitive loads (like capacitor banks) cause the current to lead the voltage.

For example, if you need 1 kW of real power at a power factor of 1.0, you only need 1 kVA of apparent power. But if your power factor is 0.2, you'd need to transfer 5 kVA just to get that same 1 kW of work! That extra power still has to be generated and transmitted, which causes extra losses in the system.

How We Correct It (PFC)

We usually want to get the power factor as close to 1.0 as possible. We do this through Power Factor Correction (PFC) by switching capacitor or inductor banks in and out of the circuit. For example, since most industrial loads are inductive (motors), we add capacitors locally to cancel out that effect. When we supply that reactive power right at the load, the overall apparent power is reduced.

Power companies do this to keep the grid stable, and customers do it to avoid those high penalty charges. It improves efficiency and helps with voltage regulation.

However, you can't just throw capacitors at a circuit. Switching them can cause voltage spikes or harmonic noise. If you leave them on with no load, you get "no-load losses," and in the worst case, the components can interact and cause resonance, which leads to huge overvoltage issues. That's why we use

Automatic Power Factor Correction (APFC)

These have a controller that monitors the current (usually with a current transformer) and switches capacitor blocks in steps to keep the power factor above a set point, like 0.9.

Alternatively, big plants sometimes use "synchronous condensers" (unloaded synchronous motors). By changing the field excitation, these can act like a variable capacitor to support the network voltage. Unlike static capacitors, the reactive power they provide is proportional to the voltage itself, which makes the system more stable.

Non-Linear Loads

Non-linear loads—like rectifiers in power supplies or fluorescent lamps—are a bit different. They "interrupt" the current, which creates harmonics (multiples of the base frequency). This changes the shape of the current wave from a nice sine wave to something messy.

In these cases, we talk about "Total Power Factor" or "True Power Factor," which accounts for both the phase shift and the harmonic distortion. To get an accurate reading on these, a standard multimeter won't work because it's

calibrated for perfect sine waves; you have to use a **True RMS** multimeter and a wattmeter designed for non-sinusoidal currents.

PROBLEM STATEMENT

Since the units are consistent, the power factor is by definition a [dimensionless number](#) between 0 and 1. When power factor is equal to 0, the energy flow is entirely reactive, and stored energy in the load returns to the source on each cycle. When the power factor is 1, all the energy supplied by the source is consumed by the load. Power factors are usually stated as "leading" or "lagging" to show the sign of the phase angle.

OBJECTIVE

- Arduino based project
- The project is designed to minimize penalty for industrial units using automatic power factor correction unit.
- Zero crossing detector concept is used .

CHAPTER 2
ABSTRACT
BLOCK DIAGRAM
CIRCUIT DIAGRAM
PCB SCHEMATIC
FLOWCHART
WORKING

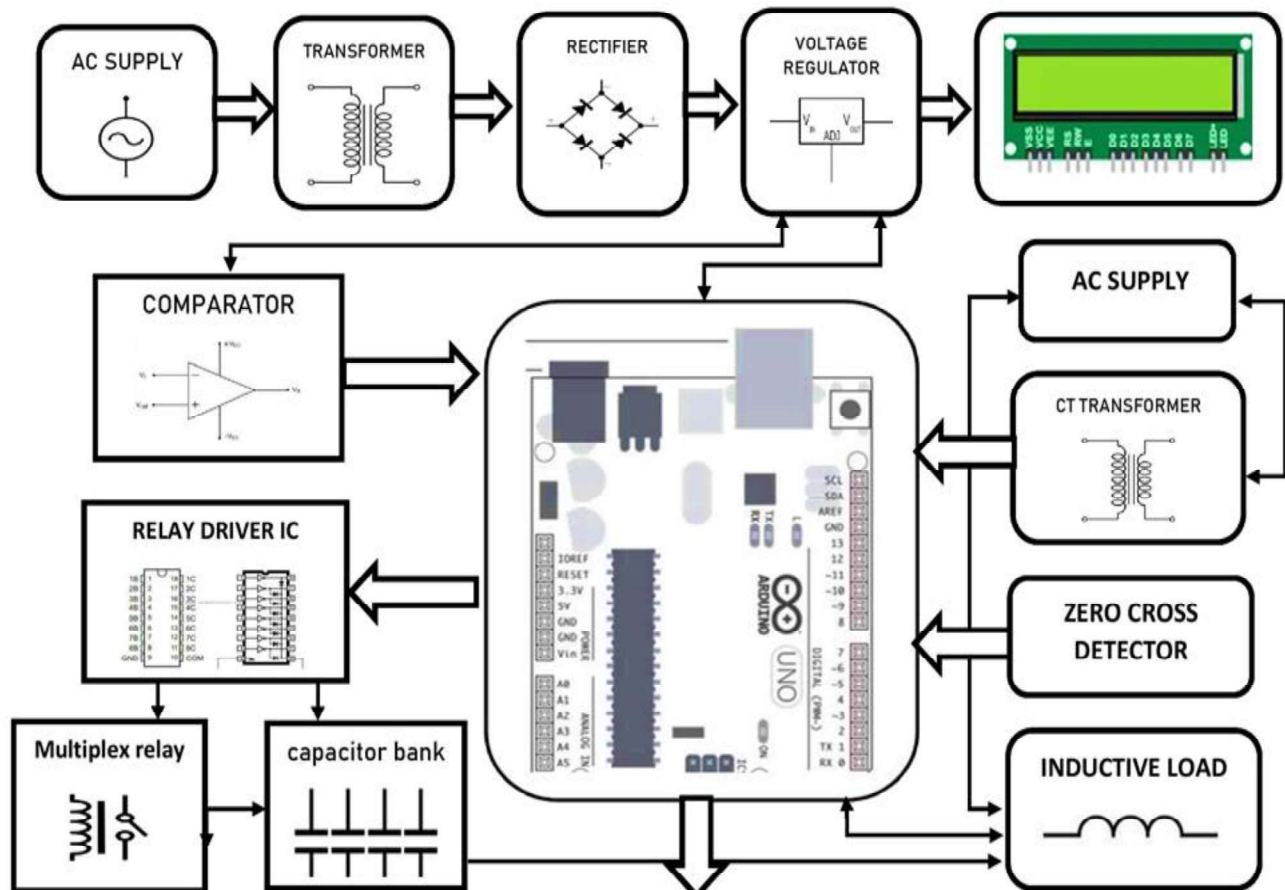
ABSTRACT

For this project, I'm using two zero-crossing detectors to pinpoint exactly when the voltage and current waves hit zero. The main goal is to build an automatic power factor correction unit that can help industrial plants avoid utility penalties. I chose an Arduino-based controller to handle the logic for the whole system.

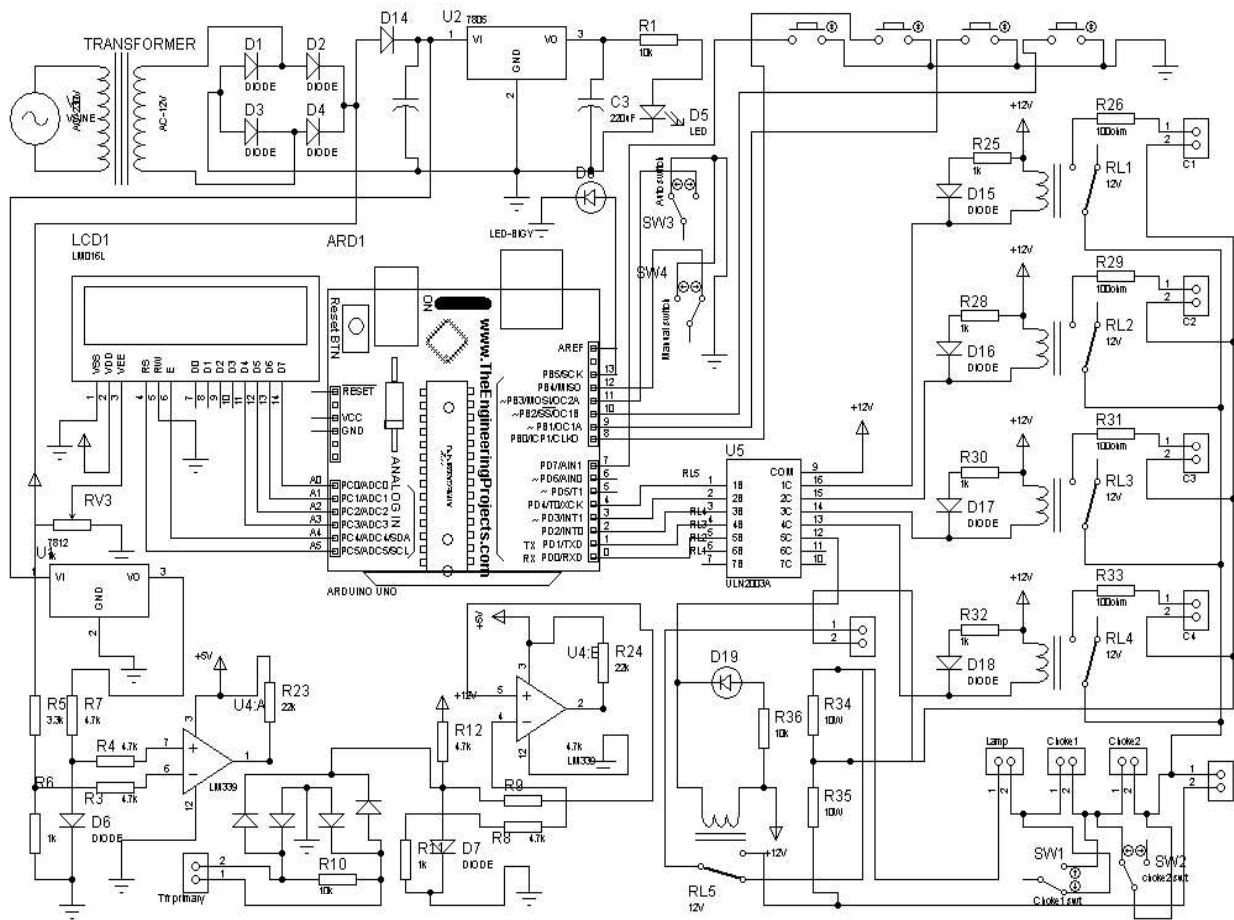
To get the timing right, I'm using operational amplifiers in comparator mode. These circuits generate pulses at the zero-crossing points, and the time lag between the voltage pulse and the current pulse is fed directly into two interrupt pins on the microcontroller. From there, the code calculates the phase shift and triggers the necessary relays. These relays switch a bank of shunt capacitors into the load circuit, step-by-step, until the power factor gets as close to unity as possible.

I've interfaced the capacitor bank and the relays to the Arduino using a relay driver, and I added an LCD to show the real-time lag between V and I. One way this could be improved later is by swapping the mechanical relays for thyristor-based switches. This would help avoid "contact pitting," which is a common problem with relays because of the high in-rush currents you get when switching capacitors.

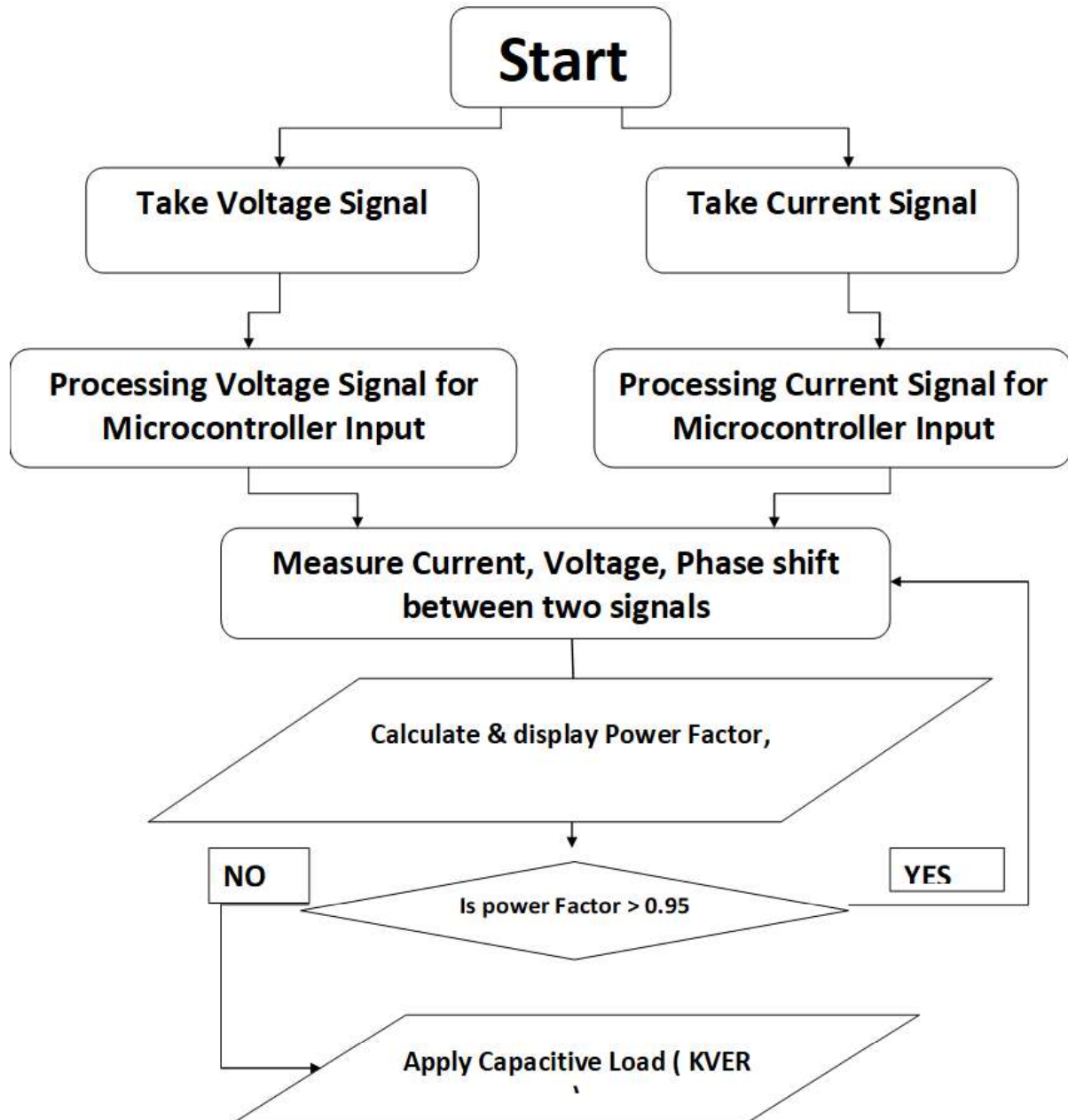
PROJECT BLOCK DIAGRAM



PROJECT CIRCUIT DIAGRAM



❖ FLOWCHARY



working of the component

Arduino Uno

I'm using an Arduino Uno as the brain of the project. It's based on the ATmega328P and has 14 digital and 6 analog pins that I can use to interface with the rest of my circuit. I programmed it using the Arduino IDE, and it can be powered either through a USB cable or a 9V battery. Basically, it takes the signals from the zero-crossing detectors, processes the timing, and then sends commands to the relays and the LCD.

Relays

The relays act as electrically operated switches. I'm using them to isolate the control circuit from the high-power side and to switch the capacitor banks in or out depending on what the Arduino decides.

Capacitor Bank

The capacitor bank is just a group of identical capacitors wired together. I'm using them in this setup to counteract the lagging power factor caused by inductive loads.

Current Transformer (CT) and Potential Transformer (PT)

These are instrument transformers used to scale down the signals to a safe level. The CT reduces the AC current to a proportional value that's easier to handle, while the PT steps down the high voltage to a lower, safer limit. This way, we can actually measure the signals without frying the components.

Zero Crossing Detector (ZCD)

The ZCD is a crucial part of the timing logic. It basically takes the sinusoidal waves and converts them into square waves. The output of the Op-Amp toggles whenever the sine wave crosses zero, which allows the Arduino to accurately measure the time difference between the V and I peaks.

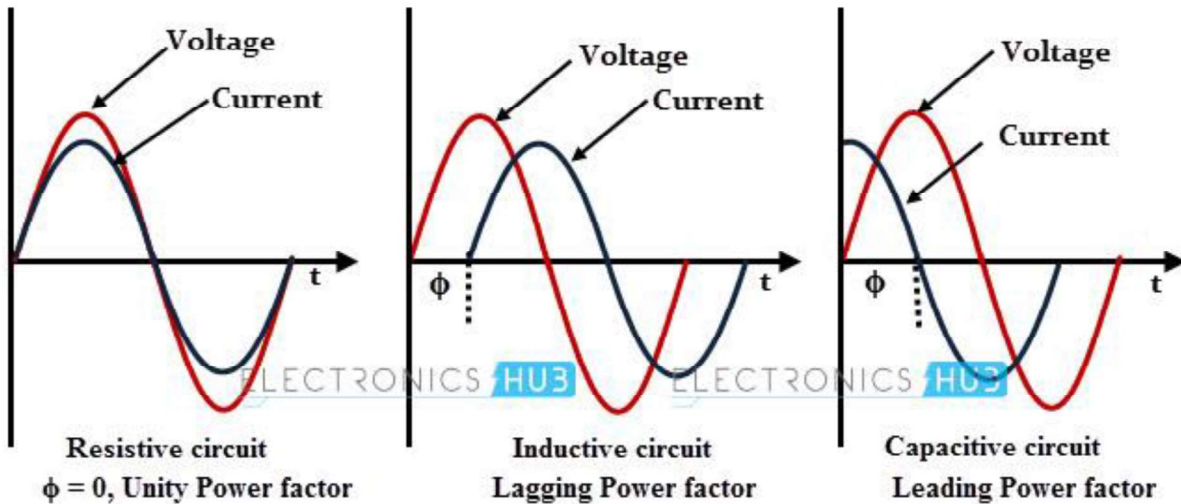
Liquid Crystal Display (LCD)

I'm using a standard 16x2 LCD module to show the power factor values. I chose this over simple LEDs or seven-segment displays because it's way easier to program and can display actual text and numbers clearly. It's a cheap and effective way to get real-time feedback from the system.

❖ Main working

The voltage or current waveform has a sinusoidal waveform with a wave cycle of 360 degrees which when converted into time seconds it is 20 milliseconds (Frequency = 50Hz, $T=1/F = 1/50$). When the load is applied the current waveforms changes its every zero-crossing position depending on the characteristics of the load. Irrespective of load the time period is similar but the

every zero crossing of the current waveform changes whereas the voltage waveform remains in the same position. So the voltage is used as the reference waveform to determine the time variation or time at which current waveform starts. The difference in the every zero crossing of the voltage and current waveforms is known as phase angle difference. This phase angle difference is used to calculate the power factor as the value of the power factor is the cosine of the Phase angle difference between voltage and current.



As mentioned above one cycle of the waveform is 20 milliseconds. The sinusoidal waveform contains a positive and negative half cycle. The waveform touches zero mark and enters negative cycle or positive cycle for every half cycle i.e. at 180 degrees which when converted into seconds it is 10 milliseconds, this process is known as zero crossing. But the zero crossing remains the same at 10 milliseconds. So when load is applied there is a difference between the zero crossing of voltage and current waveforms. The difference in their time of zero crossing is noted and converted into degrees. This value is known as phase angle difference. The power factor is the cosine of the phase angle difference of the voltage and current waveforms. So, the obtained phase angle difference gives us the value of power factor. The power for the Arduino and other components is supplied by a 5V DC supply. The two Op-Amps convert the sinusoidal signal into square signal.

CHAPTER NO 3

HARDWARE REQUIREMENTS

HARDWARE DESCRIPTION

HARDWARE REQUIREMENTS

HARDWARE COMPONENTS:

1. TRANSFORMER (230 – 12 V AC)
2. VOLTAGE REGULATOR
3. RECTIFIER
4. ULN2003
5. ARDUINO UNO
6. RELAY
7. RELAY DRIVER
8. PUSH BUTTONS
9. LCD
10. LM339
11. CURRENT TRANSFORMER
12. INDUCTIVE LOAD
13. SHUNT CAPACITOR
14. LED
15. 1N4007 / 1N4148
16. RESISTOR
17. CAPACITOR

SOFTWARE REQUIREMENT

1. Arduino ide
2. LTspice

V_s = secondary (output) voltage

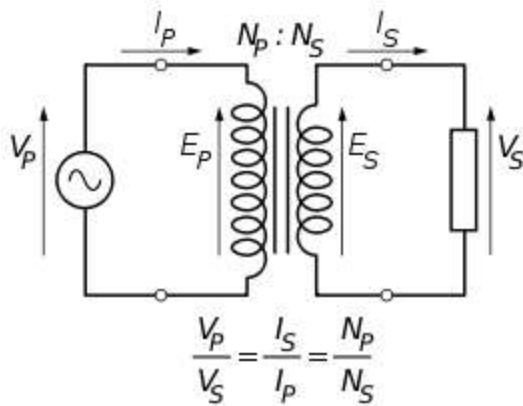
N_p = number of turns on primary coil

N_s = number of turns on secondary coil

I_p = primary (input) current

I_s = secondary (output) current.

Ideal power equation



The ideal transformer as a circuit element

If the secondary coil is attached to a load that allows current to flow, electrical power is transmitted from the primary circuit to the secondary circuit. Ideally, the transformer is perfectly efficient; all the incoming energy is transformed from the primary circuit to the magnetic field and into the secondary circuit. If this condition is met, the incoming electric power must equal the outgoing power:

$$P_{\text{incoming}} = I_p V_p = P_{\text{outgoing}} = I_s V_s,$$

giving the ideal transformer equation

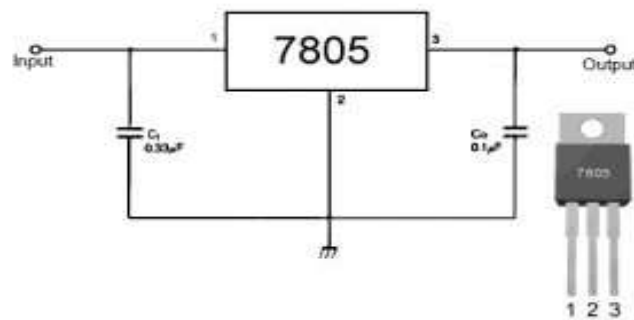
$$\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s}.$$

Transformers normally have high efficiency, so this formula is a reasonable approximation.

VOLTAGE REGULATOR 7805

Features

- Output Current up to 1A.
- Output Voltages of 5, 6, 8, 9, 10, 12, 15, 18, 24V.
- Thermal Overload Protection.
- Short Circuit Protection.
- Output Transistor Safe Operating Area Protection.



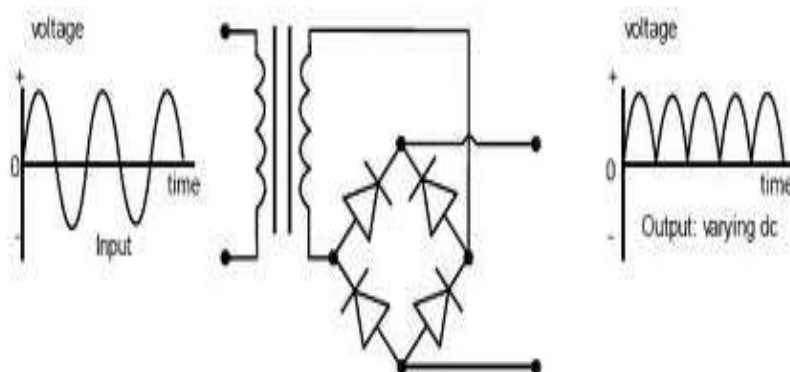
Description

This series of three-terminal positive regulators are available in the TO-220/D-PAK package and with several fixed output voltages, making them useful in a Wide range of applications. Each type employs internal current limiting, thermal shutdown and safe operating area protection, making it essentially indestructible. If adequate heat sinking is provided, they can deliver over 1A output Current. Although designed primarily as fixed voltage regulators, these devices can be used with external components to obtain adjustable voltages and currents.



RECTIFIER

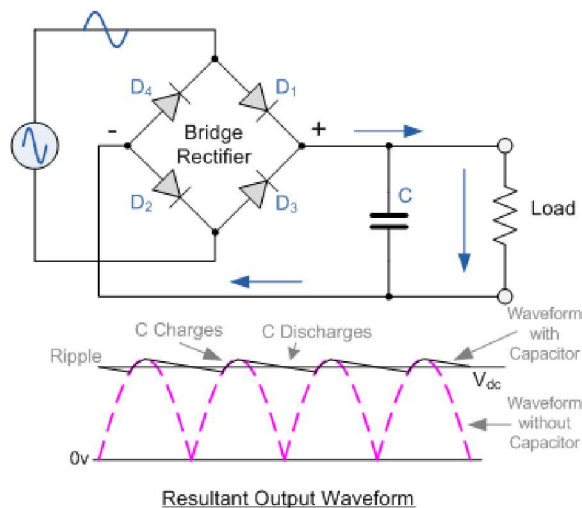
A rectifier is an electrical device that converts alternating current (AC), which periodically reverses direction, to direct current (DC), current that flows in only one direction, a process known as rectification. Rectifiers have many uses including as components of power supplies and as detectors of radio signals. Rectifiers may be made of solid state diodes, vacuum tube diodes, mercury arc valves, and other components. The output from the transformer is fed to the rectifier. It converts A.C. into pulsating D.C. The rectifier may be a half wave or a full wave rectifier. In this project, a bridge rectifier is used because of its merits like good stability and full wave rectification. In positive half cycle only two diodes (1 set of parallel diodes) will conduct, in negative half cycle remaining two diodes will conduct and they will conduct only in forward bias only.



FILTER

Capacitive filter is used in this project. It removes the ripples from the output of rectifier and smoothens the D.C. Output received from this filter is constant until the mains voltage and load is maintained constant. However, if either of the two is varied, D.C. voltage received at this point changes. Therefore a regulator is applied at the output stage.

The simple capacitor filter is the most basic type of power supply filter. The use of this filter is very limited. It is sometimes used on extremely high-voltage, low-current power supplies for cathode-ray and similar electron tubes that require very little load current from the supply. This filter is also used in circuits where the power-supply ripple frequency is not critical and can be relatively high. Below figure can show how the capacitor charges and discharges.



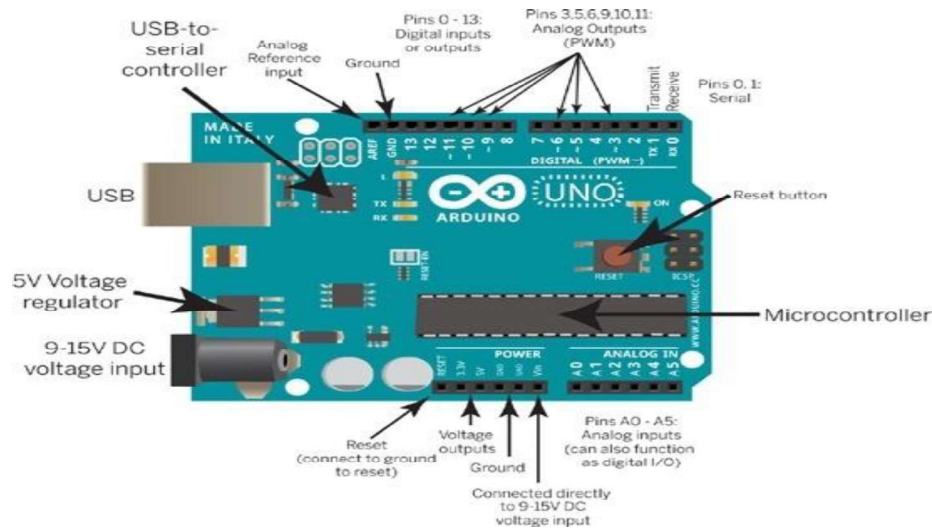
ARDUINO

INTRODUCTION:

Arduino is an open-source platform used for building electronics projects. Arduino consists of both a physical programmable circuit board (often referred to as a microcontroller) and a piece of software, or IDE (Integrated Development Environment) that runs on your computer, used to write and upload computer code to the physical board.

The Arduino platform has become quite popular with people just starting out with electronics, and for good reason. Unlike most previous programmable circuit boards, the

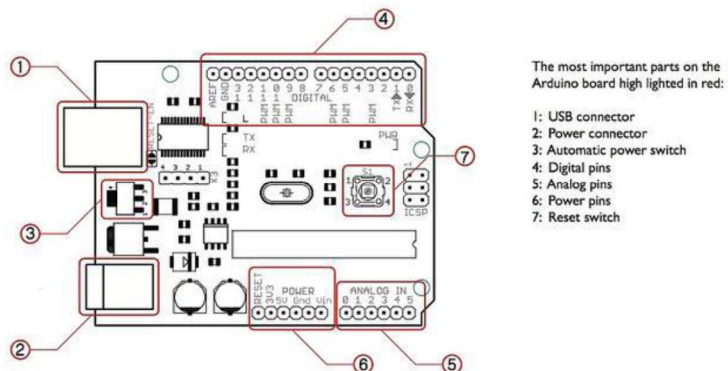
Arduino does not need a separate piece of hardware (called a programmer) in order to load new code onto the board – you can simply use a USB cable. Additionally, the Arduino IDE uses a simplified version of C++, making it easier to learn to program. Finally, Arduino provides a standard form factor that breaks out the functions of the micro-controller into a more accessible pack



Arduino

Arduino Pin Diagram

A typical example of Arduino board is Arduino Uno.



Power Jack: Arduino can be power either from the pc through a USB or through external source like adaptor or a battery. It can operate on a external supply of 7 to 12V. Power can be applied externally through the pin Vin or by giving voltage reference through the IOREf pin.

Digital Inputs: It consists of 14 digital inputs/output pins, each of which provide or take up 40mA current. Some of them have special functions like pins 0 and 1, which act as Rx and Tx respectively , for serial communication, pins 2 and 3-which are external interrupts, pins 3,5,6,9,11 which provides pwm output and pin 13 where LED is connected.

Analog inputs: It has 6 analog input/output pins, each providing a resolution of 10 bits.

ARef: It provides reference to the analog inputs

Reset: It resets the microcontroller when low.

Arduino Architecture:

Arduino's processor basically uses the Harvard architecture where the program code and program data have separate memory. It consists of two memories- Program memory and the data memory. The code is stored in the flash program memory, whereas the data is stored in the data memory. The Atmega328 has 32 KB of flash memory for storing code (of which 0.5 KB is used for the bootloader), 2 KB of SRAM and 1 KB of EEPROM and operates with a clock speed of 16MHz.

Steps to program an Arduino

- Programs written in Arduino are known as sketches. A basic sketch consists of 3 parts

1. Declaration of Variables

2. Initialization: It is written in the setup () function.

3. Control code: It is written in the loop () function.

- The sketch is saved with .ino extension. Any operations like verifying, opening a sketch, saving a sketch can be done using the buttons on the toolbar or using the tool menu.
- The sketch should be stored in the sketchbook directory.
- Chose the proper board from the tools menu and the serial port numbers.
- Click on the upload button or chose upload from the tools menu. Thus the code is uploaded by the bootloader onto the microcontroller.

Few of basic Adruino functions are:

- **digitalRead(pin):** Reads the digital value at the given pin.
- **digitalWrite(pin, value):** Writes the digital value to the given pin
- **pinMode(pin, mode):** Sets the pin to input or output mode.
- **analogRead(pin):** Reads and returns the value.
- **analogWrite(pin, value):** Writes the value to that pin.
- **serial.begin(baud rate):** Sets the beginning of serial communication by setting the bit rate.

Why Arduino?

- **Inexpensive** - Arduino boards are relatively inexpensive compared to other microcontroller platforms. The least expensive version of the Arduino module can be assembled by hand, and even the pre-assembled Arduino modules cost less than \$50
- **Cross-platform** - The Arduino Software (IDE) runs on Windows, Macintosh OSX, and Linux operating systems. Most microcontroller systems are limited to Windows.
- **Simple, clear programming environment** - The Arduino Software (IDE) is easy-to-use for beginners, yet flexible enough for advanced users to take advantage of as well. For teachers, it's conveniently based on the Processing programming environment, so students learning to program in that environment will be familiar with how the Arduino IDE works.
- **Open source and extensible software** - The Arduino software is published as open source tools, available for extension by experienced programmers. The language can be expanded through C++ libraries, and people wanting to understand the technical details can make the leap from Arduino to the AVR C programming language on which it's based. Similarly, you can add AVR-C code directly into your Arduino programs if you want to.
- **Open source and extensible hardware** - The plans of the Arduino boards are published under a Creative Commons license, so experienced circuit designers can make their own version of the module, extending it and improving it. Even relatively inexperienced users can build the breadboard version of the module in order to understand how it works.

RELAY

A relay is an electrically operated switch. Many relays use an electromagnet to operate a switching mechanism mechanically, but other operating principles are also used. Relays are used where it is necessary to control a circuit by a low-power signal (with complete electrical isolation between control and controlled circuits), or where several circuits must be controlled by one signal.

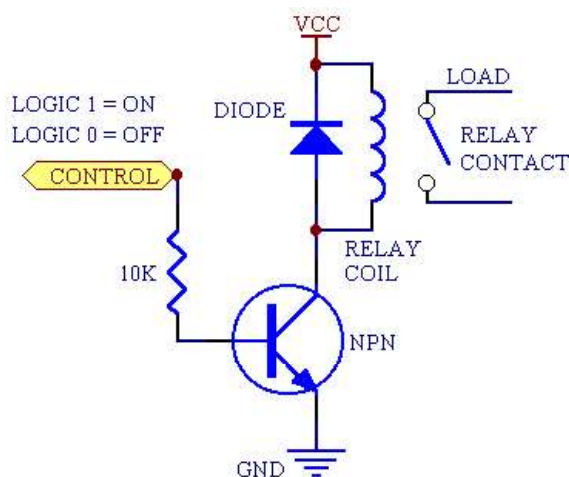


A relay is an electrically operated switch. Current flowing through the coil of the relay creates a magnetic field which attracts a lever and changes the switch contacts. The coil current can be on or off so relays have two switch positions and most have double throw (changeover) switch contacts as shown in the diagram.

Relays allow one circuit to switch a second circuit which can be completely separate from the first. For example a low voltage battery circuit can use a relay to switch a 230V AC mains circuit. There is no electrical connection inside the relay between the two circuits; the link is magnetic and mechanical.

The coil of a relay passes a relatively large current, typically 30mA for a 12V relay, but it can be as much as 100mA for relays designed to operate from lower voltages. Most ICs (chips) cannot provide this current and a transistor is usually used to amplify the small IC current to the larger value required for the relay coil.

The figure shows a relay with its coil and switch contacts. You can see a lever on the left being attracted by magnetism when the coil is switched on. This lever moves the switch contacts.



There is one set of contacts (SPDT) in the foreground and another behind them, making the relay DPDT.

The relay's switch connections are usually labelled COM, NC and NO:

- COM = Common, always connect to this; it is the moving part of the switch.
- NC = Normally Closed, COM is connected to this when the relay coil is off.
- NO = Normally Open, COM is connected to this when the relay coil is on.

ULN2003 RELAY DRIVER:

ULN2003 is a high voltage and high current Darlington transistor array.

DESCRIPTION:

The ULN2003 is a monolithic high voltage and high current Darlington transistor arrays. It consists of seven NPN Darlington pairs that feature high-voltage outputs with common-cathode Clamp diode for switching inductive loads. The collector-current rating of a single Darlington pair is 500mA. The Darlington pairs may be paralleled for higher current capability. Applications include relay drivers, hammer drivers, lamp drivers, display drivers (LED gas discharge), line drivers, and logic buffers.

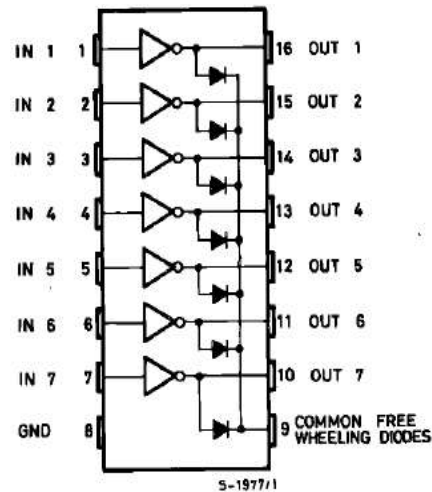
The ULN2003 has a 2.7kW series base resistor for each Darlington pair for operation directly with TTL or 5V CMOS devices.

Fig: ULN 2003

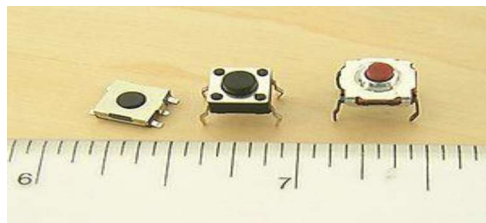


Pin no.:16

- Temperature, Operating Range:-20°C to
- Transistor Polarity: NPN
- Temp, Op. Max:85°C
- Channels, No. of:7
- Current, Output Max:500mA
- Transistor Type: Power Darlington
- Voltage, Input Max:5V
- Voltage, Output Max:50V



PUSH BUTTONS



A push-button (also spelled pushbutton) or simply button is a simple switch mechanism for controlling some aspect of a machine or a process. Buttons are typically made out of hard material, usually plastic or metal. The surface is usually flat or shaped to accommodate the human finger or hand, so as to be easily depressed or pushed. Buttons are most often biased switches, though even many un-biased buttons (due to their physical nature) require a spring to

return to their un-pushed state. Different people use different terms for the "pushing" of the button, such as press, depress, mash, and punch.

Push to ON button:



Fig 4.7(a): push on button

Initially the two contacts of the button are open. When the button is pressed they become connected. This makes the switching operation using the push button.

LIQUID CRYSTAL DISPLAY

This is the first interfacing example for the Parallel Port. We will start with something simple. This example doesn't use the Bi-directional feature found on newer ports, thus it should work with most, if not all Parallel Ports. It however doesn't show the use of the Status Port as an input for a 16 Character x 2 Line LCD Module to the Parallel Port. These LCD Modules are very common these days, and are quite simple to work with, as all the logic required running them is on board.

LCD Background:

Frequently, an Arduino program must interact with the outside world using input and output devices that communicate directly with a human being. One of the most common devices attached to an Arduino is an LCD display. Some of the most common LCDs connected to the Arduino are 16x2 and 20x2 displays. This means 16 characters per line by 2 lines and 20 characters per line by 2 lines, respectively.

Fortunately, a very popular standard exists which allows us to communicate with the vast majority of LCDs regardless of their manufacturer. The standard is referred to as HD44780U, which refers to the controller chip which receives data from an external source (in this case, the Arduino) and communicates directly with the LCD.

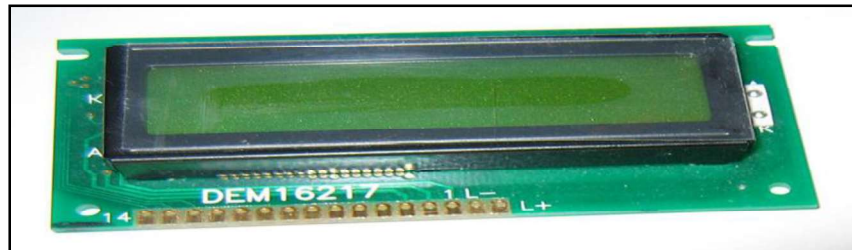
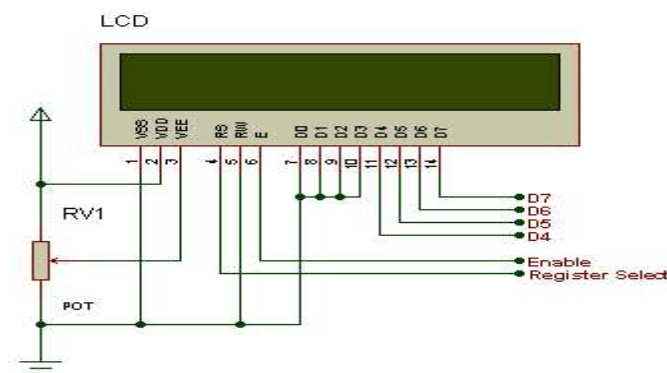


FIG 4.8: LCD

44780 LCD BACKGROUND

The 44780 standard requires 3 control lines as well as either 4 or 8 I/O lines for the data bus. The user may select whether the LCD is to operate with a 4-bit data bus or an 8-bit data bus. If a 4-bit data bus is used the LCD will require a total of 7 data lines (3 control lines plus the 4 lines for the data bus). If an 8-bit data bus is used the LCD will require a total of 11 data lines (3 control lines plus the 8 lines for the data bus).



The three control lines are referred to as **EN**, **RS**, and **RW**.

The **EN** line is called "Enable." This control line is used to tell the LCD that you are sending it data. To send data to the LCD, your program should make sure this line is low (0) and then set the other two control lines and/or put data on the data bus. When the other lines are completely ready, bring **EN** high (1) and wait for the minimum amount of time required by the LCD datasheet (this varies from LCD to LCD), and end by bringing it low (0) again.

The **RS** line is the "Register Select" line. When RS is low (0), the data is to be treated as a command or special instruction (such as clear screen, position cursor, etc.). When RS is high (1), the data being sent is text data which should be displayed on the screen. For example, to display the letter "T" on the screen you would set RS high.

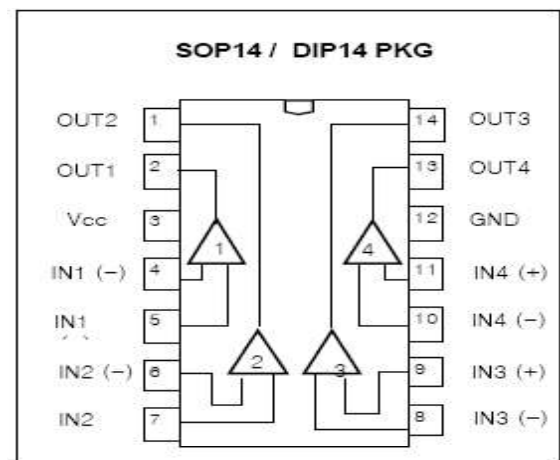
The **RW** line is the "Read/Write" control line. When RW is low (0), the information on the data bus is being written to the LCD. When RW is high (1), the program is effectively querying (or reading) the LCD.

QUAD VOLTAGE COMPARATOR LM339

DESCRIPTION

The LM339 consists of four independent precision voltage comparators, with an offset voltage specification as low as 20mV max for each comparator, which were designed specifically to operate from a single power supply over a wide range of voltages. Operation from split power supplies is also possible and the low power supply current drain is independent of the magnitude of the power supply voltage.

These comparators also have a unique characteristic in that the input common-mode voltage



ORDERING INFORMATION

Device	Package
LM339D	SOP 14
LM339N	DIP 14

range includes ground, even though they are operated from a single power supply voltage. The LM339 series was designed to directly interface with TTL and CMOS. When operated from both plus and minus power supplies, the LM339 series will directly interface with MOS logic where their low power drain is a distinct advantage over standard comparators.

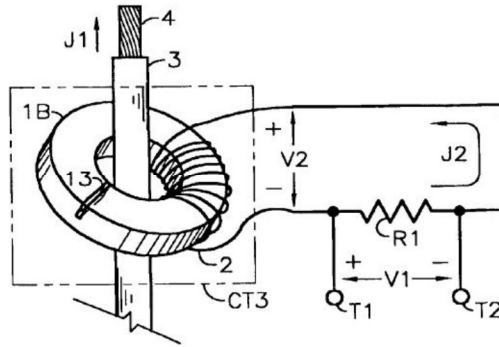
FEATURES

- Wide single supply voltage range 2.0VDC TO 36VDC or dual supplies $\pm 1.0\text{VDC}$ to $\pm 18\text{VDC}$
- Very low supply current drain (0.8mA) independent of supply voltage (1.0mW/comparator at 5.0VDC)
- Low input biasing current 25nA
- Low input offset current $\pm 5\text{nA}$ and offset voltage
- Input common-mode voltage range includes ground
- Differential input voltage range equal to the power supply voltage
- Low output 250mV at 4mA saturation voltage

APPLICATION

- A/D Converters
- Wide range VOC
- MOS clock generator
- High voltage logic gate
- Multivibrators

CURRENT TRANSFORMER



In electrical engineering, a current transformer (CT) is used for measurement of electric currents. Current transformers, together with voltage transformers (VT) (potential transformers (PT)), are known as instrument transformers. When current in a circuit is too high to directly apply to measuring instruments, a current transformer produces a reduced current accurately proportional to the current in the circuit, which can be conveniently connected to measuring and recording instruments. A current transformer also isolates the measuring instruments from what may be very high voltage in the monitored circuit. Current transformers are commonly used in metering and protective relays in the electrical power industry

INDUCTIVE LOAD



A load that is predominantly inductive, so that the alternating load current lags behind the alternating voltage of the load. Also known as lagging load. Any devices that have coils of wire in their manufacture can be classed as inductive loads. E.g. motors, solenoids and contactor coils

are a few. Example of resistive loads can be baseboard heaters, filament light bulbs, toasters and stove top elements.

SHUNT CAPACITORS



Shunt capacitor banks are used to improve the quality of the electrical supply and the efficient operation of the power system. Studies show that a flat voltage profile on the system can significantly reduce line losses. Shunt capacitor banks are relatively inexpensive and can be easily installed anywhere on the network.

LED

LEDs are semiconductor devices. Like transistors, and other diodes, LEDs are made out of silicon. What makes an LED give off light are the small amounts of chemical impurities that are added to the silicon, such as gallium, arsenide, indium, and nitride.

When current passes through the LED, it emits photons as a byproduct. Normal light bulbs produce light by heating a metal filament until it is white hot. LEDs produce photons directly and not via heat, they are far more efficient than incandescent bulbs.

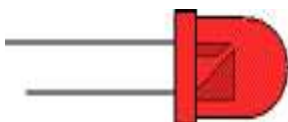


Fig 4.12(a): Typical LED



Fig 4.12(b): circuit symbol

Not long ago LEDs were only bright enough to be used as indicators on dashboards or electronic equipment. But recent advances have made LEDs bright enough to rival traditional lighting technologies. Modern LEDs can replace incandescent bulbs in almost any application.

1N4007

Diodes are used to convert AC into DC these are used as half wave rectifier or full wave rectifier. Three points must be kept in mind while using any type of diode.

1. Maximum forward current capacity
2. Maximum reverse voltage capacity
3. Maximum forward voltage capacity



1N4148

The 1N4148 is a standard small signal silicon diode used in signal processing. Its name follows the JEDEC nomenclature. The 1N4148 is generally available in a DO-35 glass package and is very useful at high frequencies with a reverse recovery time of no more than 4ns. This permits rectification and detection of radio frequency signals very effectively, as long as their amplitude is above the forward conduction threshold of silicon (around 0.7V) or the diode is biased.



Fig:1N4148 diode

Specifications:

- $V_{RRM} = 100V$ (Maximum Repetitive Reverse Voltage)
- $I_O = 200mA$ (Average Rectified Forward Current)
- $I_F = 300mA$ (DC Forward Current)
- $I_{FSM} = 1.0 A$ (Pulse Width = 1 sec), 4.0 A (Pulse Width = 1 uSec) (Non-Repetitive Peak Forward Surge Current)

RESISTORS

A resistor is a two-terminal electronic component designed to oppose an electric current by producing a voltage drop between its terminals in proportion to the current, that is, in accordance with Ohm's law:

$$V = IR$$

Resistors are used as part of electrical networks and electronic circuits. They are extremely commonplace in most electronic equipment. Practical resistors can be made of various compounds and films, as well as resistance wire (wire made of a high-resistivity alloy, such as nickel/chrome).

Units

The ohm (symbol: Ω) is the SI unit of electrical resistance, named after Georg Simon Ohm. An ohm is equivalent to a volt per ampere. Since resistors are specified and manufactured

over a very large range of values, the derived units of milliohm ($1 \text{ m}\Omega = 10^{-3} \Omega$), kilohm ($1 \text{ k}\Omega = 10^3 \Omega$), and megohm ($1 \text{ M}\Omega = 10^6 \Omega$) are also in common usage.

Variable resistors



Potentiometers

A common element in electronic devices is a three-terminal resistor with a continuously adjustable tapping point controlled by rotation of a shaft or knob. These variable resistors are known as potentiometers when all three terminals are present, since they act as a continuously adjustable voltage divider. A common example is a volume control for a radio receiver.

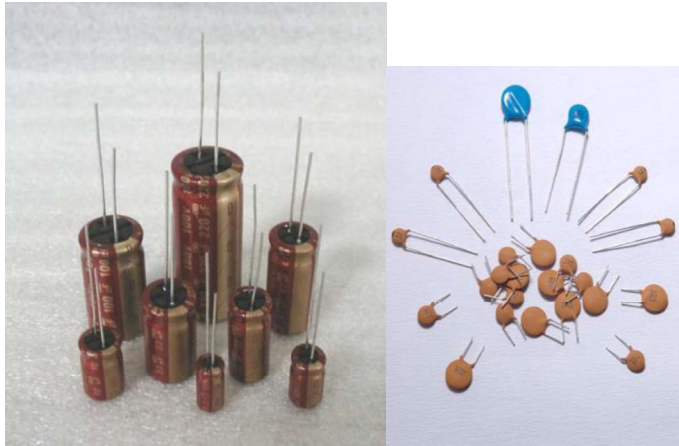
Accurate, high-resolution panel-mounted potentiometers (or "pots") have resistance elements typically wire wound on a helical mandrel, although some include a conductive-plastic resistance coating over the wire to improve resolution. These typically offer ten turns of their shafts to cover their full range. They are usually set with dials that include a simple turns counter and a graduated dial. Electronic analog computers used them in quantity for setting coefficients, and delayed-sweep oscilloscopes of recent decades included one on their panels.

CAPACITORS

A capacitor or condenser is a passive electronic component consisting of a pair of conductors separated by a dielectric. When a voltage potential difference exists between the conductors, an electric field is present in the dielectric. This field stores energy and produces a mechanical force

between the plates. The effect is greatest between wide, flat, parallel, narrowly separated conductors.

An ideal capacitor is characterized by a single constant value, capacitance, which is measured in farads. This is the ratio of the electric charge on each conductor to the potential difference between them. In practice, the dielectric between the plates passes a small amount of leakage current. The conductors and leads introduce an equivalent series resistance and the dielectric has an electric field strength limit resulting in a breakdown voltage.



A capacitor (formerly known as condenser) is a device for storing electric charge. The forms of practical capacitors vary widely, but all contain at least two conductors separated by a non-conductor. Capacitors used as parts of electrical systems, for example, consist of metal foils separated by a layer of insulating film.

Capacitors are widely used in electronic circuits for blocking direct current while allowing alternating current to pass, in filter networks, for smoothing the output of power supplies, in the resonant circuits that tune radios to particular frequencies and for many other purposes.

The capacitor is a reasonably general model for electric fields within electric circuits. An ideal capacitor is wholly characterized by a constant capacitance C , defined as the ratio of charge $\pm Q$ on each conductor to the voltage V between them:

$$C = \frac{Q}{V}$$

Sometimes charge build-up affects the capacitor mechanically, causing its capacitance to vary. In this case, capacitance is defined in terms of incremental changes:

$$C = \frac{dq}{dv}$$

Energy storage

Work must be done by an external influence to "move" charge between the conductors in a capacitor. When the external influence is removed the charge separation persists in the electric field and energy is stored to be released when the charge is allowed to return to its equilibrium position. The work done in establishing the electric field, and hence the amount of energy stored, is given by:

$$W = \int_{q=0}^Q V dq = \int_{q=0}^Q \frac{q}{C} dq = \frac{1}{2} \frac{Q^2}{C} = \frac{1}{2} CV^2 = \frac{1}{2} VQ.$$

CHAPTER NO 4

ADVANTAGES

DISADVANTAGES

APPLICATION

ADVANTAGES

- Reactive power decreases
- Efficiency of supply system and apparatus increases.
- The electrical consumption tariffs depend on power factor.
- Avoid poor voltage regulation
- Overloading is avoided
- Copper loss decreases
- Transmission loss decreases
- Improved voltage control
- Efficiency of supply system and apparatus increases

APPLICATION

- Power factor improvement **reduces load current.**
- Power factor correction reduces KVA loading of electrical equipment such as transformers and alternators which may relieve an overloaded
- system or release capacity for additional growth of load. Power factor correction improves plant efficiency.

CHAPTER NO 5

CONCLUSIO AND SCOPE

CONCLUSION

it can be concluded that power factor correction techniques can be applied to the industries,
power systems and It can be concluded that power factor correction techniques can be applied to

the industries, power systems and also households to make them stable and due to that the system becomes stable and efficiency of the system as well as the apparatus increases. The use of microcontroller reduces the costs. Due to use of microcontroller multiple parameters can be controlled and the use of extra hard wares such as timer, RAM, ROM and input output ports reduces. Care should be taken for overcorrection otherwise the voltage and current becomes more due to which the power system or machine becomes unstable and the life of capacitor banks reduces.

FUTURE SCOPE

The automatic power factor correction using capacitive load banks is a very efficient as it reduces the cost by decreasing the power drawn from the supply

Bibliography

1. Alexander, C. K., and Sadiku, M. N. O., *Fundamentals of Electric Circuits*, 7th ed. McGraw-Hill Education, 2021.
2. Boylestad, R. L., *Introductory Circuit Analysis*, 13th ed. Pearson Education, 2015.
3. Arduino CC, *Arduino Language Reference*, 2026. [Online]. Available: <https://www.arduino.cc/reference/en/>
4. Hitachi, *HD44780U Dot Matrix Liquid Crystal Display Controller Datasheet*, 1999.
5. Theraja, B. L., and Theraja, A. K., *A Textbook of Electrical Technology*, S. Chand and Company Ltd, 2005.